Field Hours

CSCE 624 Sketch Recognition

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Resample the ink before you compare

You do not classify the raw sensor stream. A stroke arrives as samples of x, y, and time, and a slow writer can emit 200 points where a fast writer emits 40. You resample by arc length to a fixed list of 64 points so both marks share a length. Spacing by time would treat a pause as a long side. Spacing by distance does not.

You then subtract the centroid so the cloud sits on the origin, and you scale the bounding box to a square of side 1. A circle 80 pixels across and a circle 20 pixels across then occupy the same frame. Leave rotation alone for now: an upward arrow and a rightward arrow are different lists until a later search tries angles. Keep the 64 time stamps in a side array, because shape and speed are different evidence.

Ink normalized before any score.

Sample x, y, t -> Resample to 64 -> Center on the centroid -> Scale the box to 1

A pause must not become an edge

1. Four samples arrive at times 0, 10, 20, and 500 milliseconds. The last gap is 480 milliseconds of waiting, not a long side of the mark.

2. Arc-length resampling to 4 points spaces those points by distance. The 480 millisecond gap does not stretch the shape.

3. You still store the times 0, 10, 20, and 500 beside the 4 points. A later speed feature can see the pause that the shape channel dropped.

Problem 1. Two writers draw the same square. One sensor emits 40 points and the other emits 200. Why is the raw list a bad feature vector?

Answer. The lengths differ, so you cannot pair coordinates. Resampling both strokes to 64 points by arc length makes the vectors the same length without treating a pause as extra shape.

Problem 2. You scale by the bounding box before you remove one outlier 30 pixels off the stroke. What goes wrong?

Answer. The outlier sets the box, so the real stroke shrinks inside the unit square. Drop or repair that point, then center and scale. A bad box warps every later distance.

Lecture 2 · 90 minutes

Features you can check by hand

You compress the 64-point stroke into a few numbers a person can audit. Divide the straight distance from the

first point to the last by the path length. A closed box lands near 0, and a straight stroke lands near 1. Count a corner wherever the turning angle exceeds 45 degrees. A rectangle often shows 4 corners, and a clean circle shows 0. These checks are not the whole recognizer.

You also store the box aspect, width divided by height, and the mean absolute turn over the 64 samples. An ellipse twice as wide as it is tall has aspect 2. Use the features as a gate before any template distance. If you asked for a circle and you already counted 4 corners, you can reject the stroke in one test. A gate on 4 corners will also reject a sloppy circle, so you fit the cutoff on labeled ink.

Four numbers you can recompute.

Closedness near 0 or 1

Corners past 45 degrees

Box width over height

Mean absolute turn

One rectangle, counted by hand

1. The path length is 40 and the gap from the first point to the last is 2. Closedness is 2 divided by 40, so 0.05, under a cutoff of 0.1, so you call the stroke closed.

2. Interior turning angles are 10, 80, 20, and 85 degrees. With a threshold of 45 degrees you count 2 corners from this list, at 80 and at 85.

3. The box is 30 pixels wide and 15 tall, so the aspect is 2. A circle gate that requires 0 corners and aspect near 1 rejects this stroke before any template runs.

Problem 1. Path length 40, start-to-end gap 2, cutoff 0.1. Is the stroke closed, and what is the ratio?

Answer. Yes. The ratio is 2 divided by 40, which is 0.05, and 0.05 is under 0.1.

Problem 2. Why fit the corner cutoff on labeled ink instead of leaving it at 45 degrees forever?

Answer. A gate at 4 corners also rejects a sloppy circle that really is a circle. The cutoff is part of the recognizer, so you choose it on strokes whose labels you already know.

Lecture 3 · 90 minutes

A geometric template distance

You score a candidate against each template only after both strokes are resampled to 64 points, centered, and scaled to the unit square. The geometric method first rotates by the indicative angle, the direction from the centroid to the first point, which removes most of the rotation in one step. A leftover error of a few degrees is reduced by golden-section search over an interval of plus or minus 45 degrees, using about 8 probes.

The path distance is the mean Euclidean gap between the 64 paired points. Divide that mean by half the diagonal of the unit square, about 0.707, and subtract the result from 1 to get a score. A perfect overlay scores 1. You label the stroke with the template class whose score is highest. If the best score on your held-out set sits under 0.75, return reject rather than a forced name.

Ordered match after one rotation.

1. Resample, center, and scale

2. Rotate by the indicative angle

3. Search a 45 degree interval

4. Keep the best score above 0.75

Four points, then a score

1. After scaling, the template and the candidate are paired. The four Euclidean gaps are 0, 0, 0, and 4. The mean gap is 1.

2. Half the diagonal of the unit square is about 0.707. The normalized distance is 1 divided by 0.707, about 1.41, which already exceeds 1.

3. You subtract that from 1 only when the normalized distance is below 1. Here the score would be negative, so you clamp it at 0 and this template loses. A perfect overlay, mean gap 0, scores 1.

Problem 1. The mean point gap is 0 and the half diagonal is about 0.707. What score do you report?

Answer. 1. The normalized distance is 0, and 1 minus 0 is 1. That is a perfect overlay.

Problem 2. The best template scores 0.70 and your reject threshold is 0.75. What do you return, and why not the nearest class anyway?

Answer. Reject. A forced label below 0.75 is a guess. The threshold is the rule you set so a poor overlay does not receive a name.

Lecture 4 · 90 minutes

Clouds when stroke order wanders

A symbol made of several strokes does not have a stable order. One writer draws the crossbar of a T second, and another draws it first. You drop order: resample each stroke, join the samples into one cloud of 32 points, and match that cloud to a template cloud. Direction dies with order. A clockwise trace and a counterclockwise trace can land on the same 32 points.

Greedy matching is not the ordered geometric distance. For each of the 32 candidate points you claim the nearest template point that is still free, add the Euclidean cost, and move on. The sum divided by 32 is the cloud distance. You pay when two classes share a silhouette but not a path, such as arrows with opposite heads. Then keep an ordered score too, and let a margin of 0.1 decide which one you trust.

Order is information you sometimes drop.

Ordered path

64 paired points

Direction is kept

Stroke order matters

Point cloud

32 greedy pairs

Direction is dropped

Order may vary

A three-point greedy match

1. The candidate points must match unused template points. Costs from the first candidate are 1 to template A and 4 to template B. You take A.

2. The second candidate sees only B, at cost 2. The third sees the last template at cost 3. The sum is 1 plus 2 plus 3, so 6.

3. Divide by 3 points, not by 32, in this toy: the cloud distance is 2. On the real recognizer you divide the sum by 32. A second class with sum 3 would win.

Problem 1. In the three-point toy the costs are 1, 2, and 3. What is the cloud distance?

Answer. 2. The sum is 6 and you divide by 3 points.

Problem 2. Two arrow classes share a silhouette but the heads point opposite ways. Why is the cloud score not enough?

Answer. Greedy matching ignores path direction, so both arrows can land on the same 32 points. You keep an ordered score beside the cloud score and let a margin of 0.1 decide which distance you trust.

Lecture 5 · 90 minutes

Speed and pauses as evidence

Speed is what the geometric normalizer throws away when it resamples by arc length. Between raw samples you divide distance by the time gap. A flick and a careful trace can share a shape and still differ by a factor of 4 in median speed. Count a pause when a gap exceeds 150 milliseconds. A tap is one short stroke, high speed, and 0 pauses. A box drawn while thinking may show 2 pauses at corners.

Keep timing as its own score. Build a histogram of 4 speed bins and normalize so the bins sum to 1. The sum of absolute bin differences is 0 on a match and at most 2 if the histograms are disjoint. If the best geometric score and the runner-up differ by less than 0.05, you may let the timing distance break the tie. Do not add that distance to a geometric score: the units differ.

Timing stays in its own units.

Measure each time gap -> Fill 4 speed bins -> Normalize the bins to 1 -> Compare histograms

One pause and one speed

1. Successive gaps are 20, 20, 20, and 200 milliseconds. The pause threshold is 150, so you count 1 pause, the last gap.

2. The first segment covers 10 pixels in 20 milliseconds, so the speed is 0.5 pixels per millisecond. That value falls in one of the 4 bins.

3. Two normalized histograms differ by absolute amounts 0.1, 0.1, 0.0, and 0.2. The timing distance is 0.4. You do not add 0.4 to a geometric score near 0.9.

Problem 1. Gaps are 20, 20, 20, and 200 milliseconds, and the pause cutoff is 150. How many pauses do you count?

Answer. 1. Only the gap of 200 exceeds 150.

Problem 2. Geometric scores are 0.91 and 0.88. May timing break the tie, and may you add the raw timing distance to 0.91?

Answer. Timing may break the tie, because the geometric margin is 0.03, which is under 0.05. You still may not add the raw timing distance to 0.91. The units differ, so you compare the histograms on their own scale.

Lecture 6 · 90 minutes

A grid of pixels beside the path

Some marks are easier as pixels than as paths. You paint the centered stroke on a 24 by 24 grid, 576 cells, and turn a cell on if any point lands in it. A thin stroke might light 40 cells, so the density is 40 divided by 576. A cell stores no stroke id, so order among 2 strokes is already gone. A wide pen changes the density, and you should fix the pen width before you compare grids.

You can score two grids by Hamming distance, the count of cells that disagree. Identical grids score 0. Missing a bar of 10 cells costs at least 10. At 24 by 24, a hook under one cell vanishes, so do not let pixels be your only vote. Keep the ordered geometric score next to the Hamming count, and normalize later: a distance of 10 on 576 cells is not a similarity of 10.

A path and a picture see different mistakes.

Ordered path

Keeps a small hook

Uses point order

Score near 1 is good

24 by 24 grid

Drops a sub-cell hook

Forgets stroke order

Hamming 0 is perfect

Forty cells on, ten cells off

1. The grid has 576 cells and the stroke lights 40 of them. Density is 40 divided by 576, about 0.069.

2. A template disagrees on 10 cells. The Hamming distance is 10, not 10 divided by anything yet.

3. The mapped similarity is 1 minus 10 divided by 576, about 0.983. You may compare that number

with a geometric score. You may not compare the raw 10 with a score that already lives near 1.

Problem 1. 576 cells, 40 lit. What is the density as a fraction?

Answer. 40 divided by 576, which is 5 divided by 72, about 0.069.

Problem 2. Hamming distance is 10 on 576 cells. Why is 10 not the similarity you add to a geometric score?

Answer. A geometric score already sits near 1, and 10 is a count of cells. Map it first: 1 minus 10 divided by 576, about 0.983. Then the units match.

Lecture 7 · 90 minutes

Three scores, one decision

You now hold three numbers: a geometric similarity, a timing distance, and a pixel Hamming count. They are not in one unit, so you map each onto 0 to 1 before any sum. A Hamming count h on 576 cells becomes 1 minus h divided by 576. A timing distance you have capped at 2 becomes 1 minus that distance divided by 2. A weighted sum can then use weights 0.5, 0.3, and 0.2.

You fit those three weights on 50 labeled strokes per class by counting mistakes, not by preference. After the weights are fixed, reject a decision when the top two class totals differ by less than 0.05. Also reject when the winner itself is under 0.6. The first test catches a tie between circle and oval. The second catches a scribble that matches no class. Store the rejects: they are the next 10 templates you should add.

Map, weight, then maybe reject.

1. Map each score onto 0 to 1

2. Apply weights 0.5, 0.3, 0.2

3. Reject a margin under 0.05

4. Reject a winner under 0.6

One weighted total

1. Mapped scores are geometric 0.80, timing 0.50, and pixel 0.70. Weights are 0.5, 0.3, and 0.2.

2. The products are 0.40, 0.15, and 0.14. The total is 0.69.

3. A runner-up total of 0.66 differs by 0.03, which is under 0.05, so you reject even though 0.69 clears 0.6. Both tests have to pass.

Problem 1. Mapped scores 0.80, 0.50, and 0.70 with weights 0.5, 0.3, and 0.2. What is the total?

Answer. 0.69. The products are 0.40, 0.15, and 0.14.

Problem 2. The top totals are 0.69 and 0.66. The winner is above 0.6. Do you label the stroke?

Answer. No. The margin is 0.03, under 0.05, so the tie test rejects. Clearing 0.6 is not enough by itself.

Lecture 8 · 90 minutes

A recognizer you can defend

You can now take a raw stroke, resample it to 64 points, score ordered templates, score a 4-bin speed histogram, paint a 24 by 24 grid, and combine the three mapped scores with weights fit on 50 strokes per class. You can refuse the label when the top two totals differ by less than 0.05. That is a recognizer you can implement and inspect one stage at a time. It does not promise that every symbol family separates.

What you still owe is a count on ink you did not tune on. Hold out 20 strokes per class, run the full stack, and tally errors by stage. If circles and ovals collide in the ordered distance, add the aspect feature from the second lecture and refit only the weights. If pauses separate a gesture from a shape, let the timing term cast the deciding vote. You should be able to name the stage that caused a given miss.

The stack you can point at.

64 resampled points

Three mapped scores

Weights from 50 strokes

Reject or emit a label

Shape ties, timing does not

1. Both the circle class and the gesture class have geometric score 0.90. Timing scores are 0.20 for the circle and 0.80 for the gesture.

2. With equal weights 0.5 and 0.5, the circle total is 0.55 and the gesture total is 0.85.

3. The gesture wins by 0.30. The margin is above 0.05, and 0.85 is above 0.6, so you emit the gesture label. Geometry alone could not have chosen.

Problem 1. Hold out 20 strokes per class and 4 of the circle strokes are labeled oval. Where do you look first?

Answer. The ordered geometric stage, because circles and ovals collide there. Add the aspect feature and refit only the weights. Do not retune on these 20 strokes and then quote them as a test.

Problem 2. Combine the geometric score and the timing score. Both classes score 0.90 on geometry. Timing is 0.20 and 0.80. Weights are 0.5 and 0.5. Which class wins, and by how much?

Answer. The gesture class. Its total is 0.85 and the circle total is 0.55, so the margin is 0.30. You emit the gesture label because the margin clears 0.05 and the winner clears 0.6.